RoadLens: Robust Traffic Sign Classification under Common Image Corruptions

Mixed augmentation, gated model fusion and test-time adaptation, evaluated on GTSRB-C at pilot scale

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ABSTRACT

Traffic sign classifiers reach about 99.8 % accuracy on clean benchmark images, yet their accuracy under noise, blur, weather and compression is much less settled. We study how far a clean-trained classifier can be made robust to such corruptions with three ingredients: AugMix-style mixing combined with PRIME-style spectral and spatial primitives under a Jensen–Shannon consistency loss, a gated fusion of a ResNet-18 and a DeiT-S, and test-time adaptation of BatchNorm layers. We build a 64 px corruption benchmark with 14 test corruptions at 5 severities, a disjoint set of validation corruptions for model selection, and a cross-dataset test on 22 sign classes shared with the Belgian BTSD. In a pilot-scale evaluation (one seed, 40 % of the training epochs, 2,000 test images, one T4 GPU) the proposed training raises ResNet-18 corrupted accuracy from 83.63 % to 93.56 %, mostly on noise (55.1 % to 89.0 %). The learned gate did not beat a plain average of its two branches (95.23 % against 95.38 %), and TENT added 0.18 points. All models are underconfident rather than overconfident, a consequence of label smoothing. The study is not a full-scale result; we state what the pilot can and cannot support.

Keywords: traffic sign recognition; corruption robustness; data augmentation; model fusion; test-time adaptation; calibration

SCOPE OF THE EVIDENCE

All results come from the pilot profile of our pipeline: a single training seed, 40 % of the training schedule, and the first 2,000 GTSRB test images. There are therefore no standard deviations or significance tests, and differences under about half a point should be read as ties. The full three-seed experiment has not been run.

1 Introduction

Safety-critical vision systems meet images that differ from clean benchmark photographs: sensor noise at night, fog, motion blur, aggressive compression. For traffic sign recognition, the German Traffic Sign Recognition Benchmark (GTSRB) [1] is close to solved on clean data, with reported accuracies around 99.8 % [2]. The practical question is how accuracy degrades when images are corrupted, and which inexpensive training and inference measures reduce that degradation.

General-purpose corruption benchmarks and defences exist for ImageNet [3, 4, 5], but their findings do not automatically carry over to small, low-resolution, 43-class traffic sign images, and published results for corrupted GTSRB use differing corruption sets and protocols [6]. We therefore build a self-contained benchmark and test a compact set of defences against it. We ask six research questions:

1. **RQ1.** How much corruption robustness does mixed augmentation with a consistency loss add over standard training?
2. **RQ2.** Which components of that recipe account for the gain?
3. **RQ3.** Does a learned gate over a CNN and a transformer improve on simply averaging them?
4. **RQ4.** When does test-time adaptation help, and how does it depend on batch size?
5. **RQ5.** Do the conclusions transfer to a second dataset?
6. **RQ6.** How well calibrated are the resulting models?

Contributions. (i) A reproducible 64 px corruption benchmark with a validation-corruption set that is disjoint from the test corruptions, per-image seeded generation and a leave-category-out protocol for augmentation selection. (ii) A controlled component ablation from a standard ResNet-18 to fusion and adaptation. (iii) A negative result: a learned gate does not beat the unweighted average of the same two models at this scale. (iv) A measured account of when BatchNorm-based adaptation helps and when it collapses. (v) A calibration analysis showing that high ECE here reflects underconfidence induced by label smoothing. We do not propose a new augmentation: the training recipe combines existing components [4, 5].

2 Related work

Traffic sign recognition. GTSRB [1] introduced a 43-class benchmark on which convolutional networks matched or exceeded human accuracy; later networks report 99.8 % and above [2]. The Belgian BTSD [7] offers a second, differently photographed sign set with partly overlapping classes, which we use to test transfer. Robustness of GTSRB models to image corruptions has been examined [6], typically by applying ImageNet-C-style transformations.

Corruption benchmarks and augmentation. ImageNet-C [3] defines 15 corruptions at five severities and the mean corruption error (mCE). AugMix [4] mixes chains of augmentation operations and adds a Jensen–Shannon consistency loss between the clean and augmented views; PRIME [5] replaces the operation set by a few max-entropy primitives that are spectral, spatial and colour transformations. A Fourier-domain analysis of model sensitivity [8] motivates spectral perturbations. Our recipe combines the AugMix chain and loss with PRIME-style primitives.

Test-time adaptation. Re-estimating BatchNorm [9] statistics on the test distribution improves robustness [10]. Tent [11] additionally minimises prediction entropy by updating the BatchNorm scale and shift online. Both rely on batch statistics, which is the dependence we measure in RQ4.

Architectures and ensembles. Residual CNNs [12] and data-efficient vision transformers [13, 14] have different inductive biases and may fail on different corruptions, which motivates combining them; deep ensembles [15] are a strong baseline for such a combination.

Calibration. Modern networks are often overconfident [16]. Label smoothing [17] reduces overconfidence and affects what ECE measures [18], which is relevant to RQ6.

3 Benchmark

3.1 Data

GTSRB provides 39,209 training frames and 12,630 test images in 43 classes [1]. Images are resized to 64 px *before* corruption. Ten percent of training data is held out for validation, split by physical sign track (about 30 frames per sign), so near-duplicate frames cannot leak. This yields 35,219 training and 3,990 validation images. The pilot evaluates the first 2,000 test images.

BTSD [7] has 62 source classes; 22 are the same sign as a GTSRB class (2,734 images, class map verified manually). Models are never trained on BTSD. The Chinese CTSD set could not be obtained and is excluded.

3.2 Corruptions

Fourteen test corruptions in four categories, each at five severities, use constants adapted from Tiny-ImageNet-C [3] for 64 px:

- **Noise:** Gaussian, impulse, shot. **Blur:** motion, zoom, defocus.
- **Weather:** snow, fog, frost, brightness. **Digital:** contrast, pixelate, elastic transform, JPEG.

Each image receives its own seed derived from (corruption, severity, image index), making the benchmark independent of worker count and generation order. Validation uses four different corruptions (speckle noise, Gaussian blur, spatter, saturate) and is used only for early stopping and checkpoint selection. The pilot yields $14 \times 5 \times 2{,}000 = 140{,}000$ corrupted test images per dataset.

4 Method

4.1 Models

We compare SmallCNN (1.2 M parameters, from scratch), ResNet-18 [12] (11.2 M) and DeiT-S [13] (21.6 M), the latter two initialised from ImageNet weights via timm [19] and fed images upsampled from 64 to 128 px inside the model. Each has a *baseline* variant (mild colour jitter only, no horizontal flip

because left and right signs are mirror images) and, for ResNet-18 and DeiT-S, a *proposed-training* variant.

4.2 Proposed training

Each image is processed into a clean view and two mixed views. Mixing follows AugMix [4] (chains of width 3 and depth 1 to 3, Dirichlet and Beta weights with concentration 1, magnitude 3 of 10) over autocontrast, equalize, posterize, solarize, rotate, shear and translate, plus four PRIME-style primitives [5]: Fourier-basis noise, random 3×3 spectral filters, smooth spatial warps and colour curves. No training operation is one of the 14 test corruptions. The loss is

```
L = CE(f(x_clean), y) + 12 * JSD( p_clean, p_view1, p_view2 )
```

optimised with AdamW [20] (learning rate 5e-4, weight decay 0.05, one warm-up epoch then cosine decay, label smoothing 0.1 [17], gradient clipping 1.0, batch 128, mixed precision) with early stopping on validation-C accuracy (patience 4).

4.3 Gated fusion

The fusion combines the two proposed-training branches as $p(y|x) = \sum_k w_k(x)\ \mathrm{softmax}(z_k(x))$. The gate is an MLP over the branches' pooled features and probabilities, initialised to the plain average. Both branches are frozen, leaving 0.25 M of 33.1 M parameters trainable. We include the unweighted average and a confidence-weighted average of the same branches [15] as controls, so the gate's own contribution is isolated.

4.4 Test-time adaptation

Test-time BN [10] uses the statistics of each test batch. TENT [11] also updates BatchNorm scale and shift by one Adam step per batch on the prediction entropy. Models are reset before every corruption set. Both methods act on BatchNorm layers, so they apply to SmallCNN, ResNet-18 and the ResNet branch of the fusion.

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5 Experimental protocol

5.1 Metrics

Accuracy, macro-F1 and ECE with 15 equal-width bins [16]. The mean corruption error follows [3] but is normalised by the SmallCNN rather than AlexNet:

```
mCE     = 100 * mean_c [ sum_s E(c,s) / sum_s E_SmallCNN(c,s) ]
rel mCE = the same with the clean error subtracted on both sides
```

SmallCNN therefore scores 100 by construction, lower is better, and our mCE values are not comparable with ImageNet-C mCE. ECE in the tables is averaged over the 70 corruption sets; reliability diagrams pool them.

5.2 Experiments and pilot profile

Six experiments: clean accuracy (E1); all corruptions and severities with mCE and ECE (E2); component ablation (E3); leave-category-out augmentation selection (E4); BTSD transfer (E5); test-time adaptation against batch size 1, 16, 64 and 200 at severity 5 (E6). The pilot profile trains on the full 35,219 images with one seed (0), scales epochs to 40 % (SmallCNN 12, other models 6, fusion 2), tests on 2,000 images, and uses 500 images per set (100 for batch size 1) in E6.

5.3 Statistical treatment

With one seed we report no standard deviations. For binomial uncertainty from the finite test set we give Wilson 95 % half-widths for accuracy: about 0.3 to 0.5 points for clean accuracy at n = 2,000. The 140,000 corrupted samples are the same 2,000 images under 70 corruptions, so we treat n = 2,000 as a conservative bound for corrupted accuracy. Paired tests are not possible because per-image predictions were not stored.

5.4 Setup

NVIDIA Tesla T4 (15 GB, driver 580.82.07), Python 3.13, PyTorch 2.11.0 with CUDA 13.0, Google Colab. 39 unit tests pass. The pilot's six experiments and figures took 73 minutes.

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6 Results

6.1 Clean accuracy (E1)

Clean accuracy is saturated: all seven models lie within 0.8 points and train accuracy is about 100 %. At n = 2,000 one image is 0.05 points and the 95 % half-width is 0.2 to 0.5 points, so the nominal ordering is not informative.

Table 1. Clean accuracy on the first 2,000 GTSRB test images (%). CI is the Wilson 95 % half-width in points.

Model	Params (M)	Train acc.	Test acc.	±95 % CI	Macro-F1	ECE
SmallCNN	1.20	100.00	**99.60**	0.29	99.45	12.31
ResNet-18	11.20	100.00	98.80	0.49	98.03	10.39
DeiT-S	21.60	100.00	99.10	0.42	98.47	8.67
ResNet-18 (proposed training)	11.20	100.00	99.20	0.40	98.72	10.06
DeiT-S (proposed training)	21.60	99.99	99.45	0.34	99.32	8.90
Confidence-weighted ensemble	32.80	99.99	99.45	0.34	99.38	9.49
Gated fusion (proposed)	33.10	99.99	99.50	0.32	**99.48**	9.37

6.2 Corruption robustness (E2) — RQ1

Standard ResNet-18 is the weakest model under corruption (83.63 %, mCE 258.1), well below the 1.2 M-parameter SmallCNN (91.84 %). The proposed training lifts ResNet-18 by 9.93 points to 93.56 % and DeiT-S by 1.55 points to 94.85 %, which are differences well outside the conservative interval (about ±0.9 points). The three combination models are within 0.15 points of one another. Relative mCE subtracts a clean error that is already near 0.5 %, so it swings widely (53.1 to 105.5) and we do not interpret it further at this scale.

Table 2. Accuracy under corruption, averaged over 14 corruptions × 5 severities (%). mCE is normalised by SmallCNN. The CI is a conservative Wilson half-width at n = 2,000.

Model	Clean acc.	Corr. acc.	±95 % CI	mCE	Rel. mCE	Corr. F1	Clean ECE	Cor
SmallCNN	**99.60**	91.84	1.20	100.00	100.00	91.27	12.31	1
ResNet-18	98.80	83.63	1.62	258.10	328.00	81.66	10.39	
DeiT-S	99.10	93.30	1.10	140.50	127.30	92.19	8.67	
ResNet-18 (proposed training)	99.20	93.56	1.08	122.80	110.60	92.09	10.06	
DeiT-S (proposed training)	99.45	94.85	0.97	102.30	131.50	93.60	8.90	
Average ensemble	99.40	**95.38**	0.92	**84.80**	**53.10**	**94.44**	9.58	1
Confidence-weighted ensemble	99.45	95.36	0.93	84.90	69.80	94.41	9.49	
Gated fusion (proposed)	99.50	95.23	0.94	89.70	105.50	94.07	9.37	

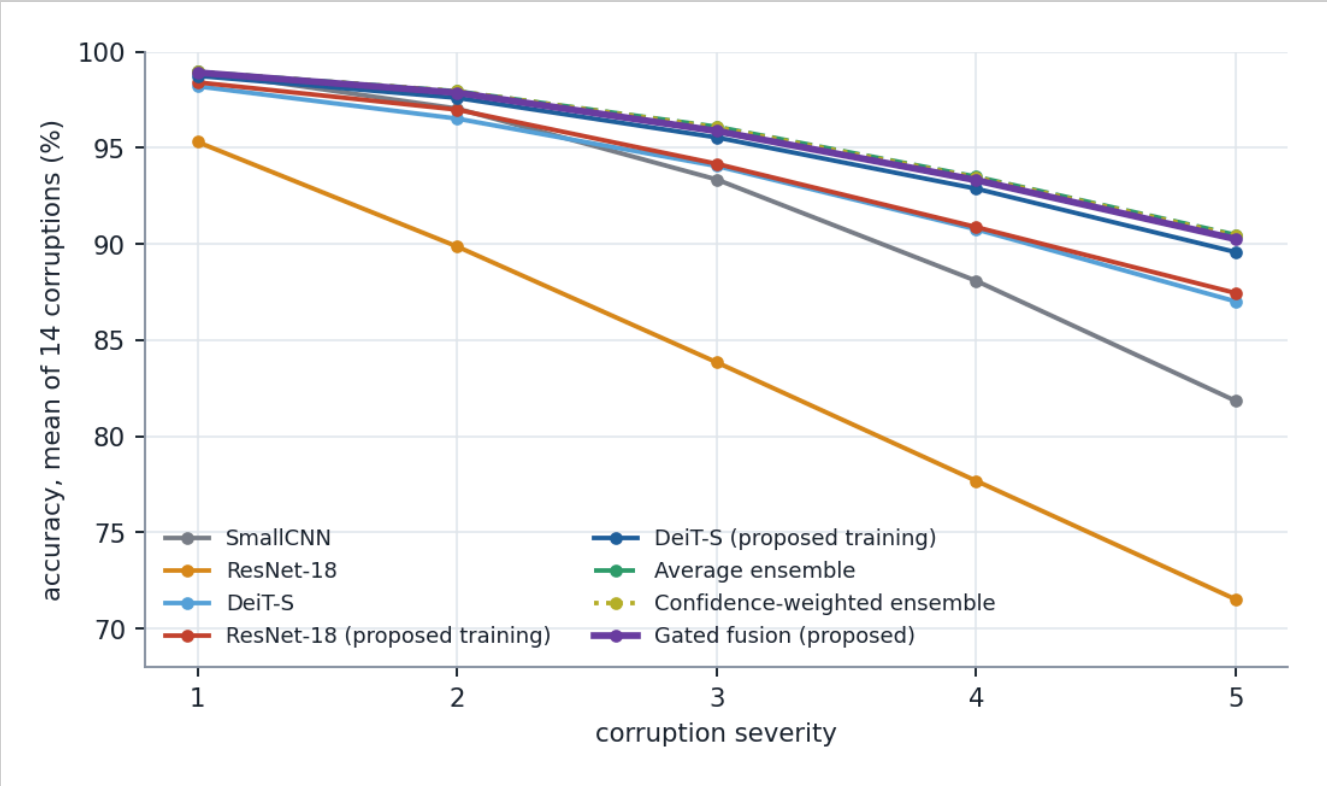


Figure 1. Mean accuracy over the 14 corruptions at each severity. Models are indistinguishable at severity 1 and spread by about 19 points at severity 5.

Table 3. Mean accuracy over the 14 corruptions at each severity (%).

Model	s1	s2	s3	s4	s5
SmallCNN	**98.96**	97.04	93.35	88.08	81.84
ResNet-18	95.30	89.86	83.84	77.68	71.51
DeiT-S	98.19	96.51	94.05	90.76	87.01
ResNet-18 (proposed training)	98.40	96.98	94.16	90.87	87.44
DeiT-S (proposed training)	98.74	97.59	95.54	92.87	89.57
Average ensemble	98.94	**97.94**	**96.09**	**93.49**	**90.47**
Confidence-weighted ensemble	98.94	**97.94**	96.06	93.46	90.44
Gated fusion (proposed)	98.91	97.86	95.87	93.31	90.25

At severity 1 every model is within 3.6 points; at severity 5 the spread is 19 points (71.5 to 90.5 %). Averaging over severities therefore understates the differences that matter at the hard end.

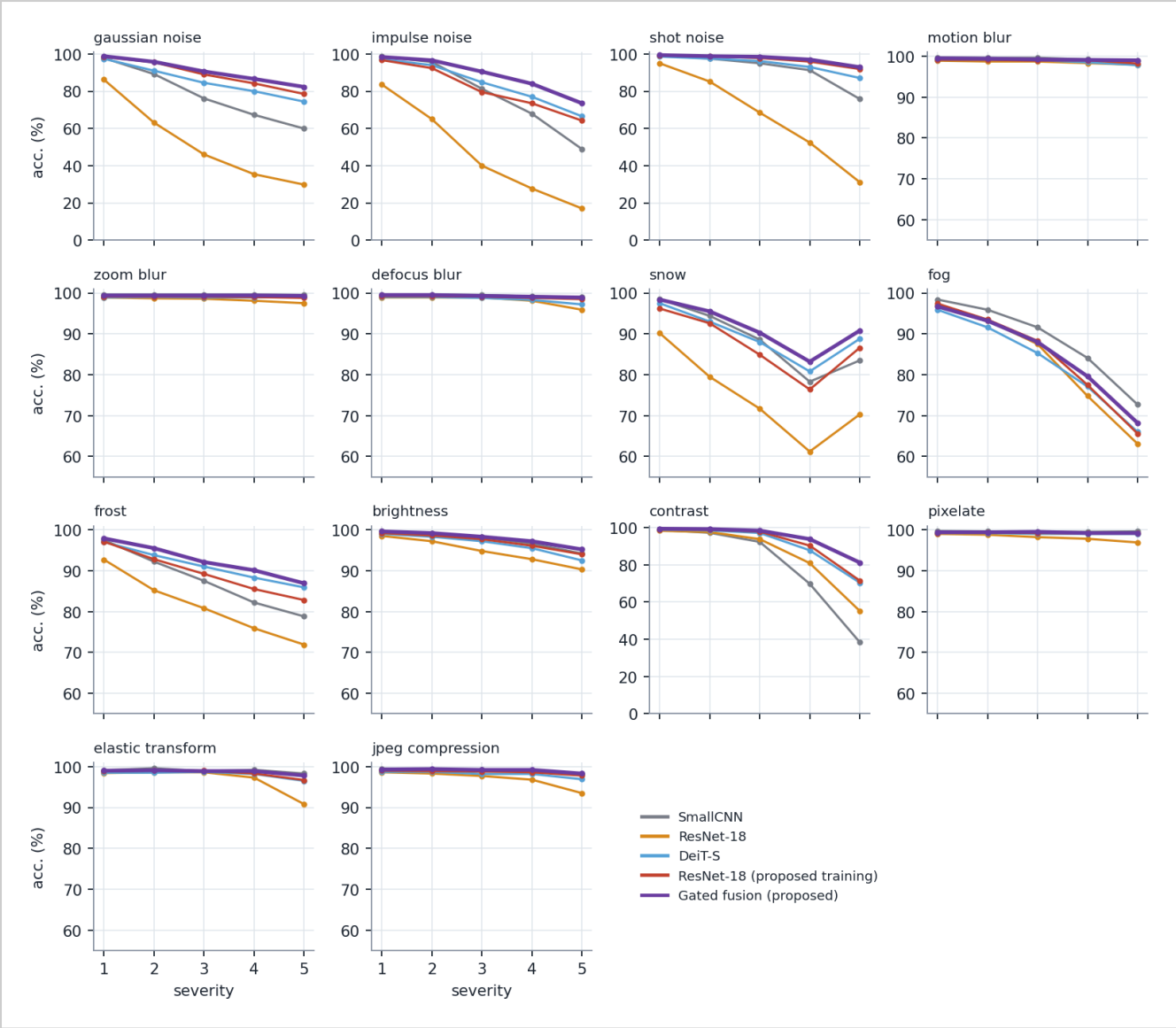


Figure 2. Accuracy against severity for each corruption (five representative models). Snow accuracy rises from severity 4 to 5 for every model, a property of that corruption ladder.

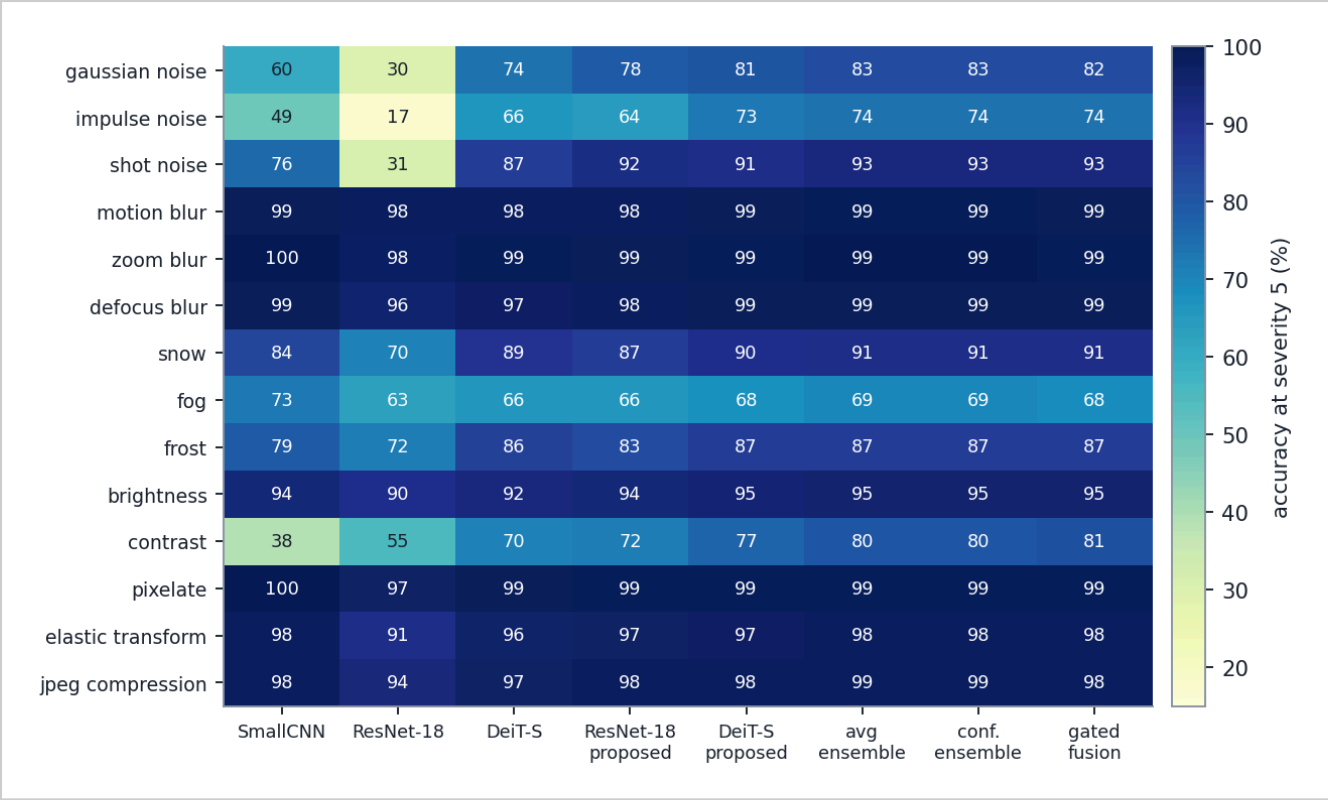


Figure 3. Severity-5 accuracy per corruption and model. Standard ResNet-18 falls to 17.0 % on impulse, 29.8 % on Gaussian and 31.1 % on shot noise.

Table 4. Accuracy by corruption category, averaged over severities (%).

Model	Noise	Blur	Weather	Digital
SmallCNN	82.77	**99.45**	90.59	94.22
ResNet-18	55.07	98.33	83.44	94.25
DeiT-S	87.94	98.71	90.12	96.46
ResNet-18 (proposed training)	89.03	98.95	89.60	96.92
DeiT-S (proposed training)	91.67	99.13	91.47	97.44
Average ensemble	**92.48**	99.31	**92.07**	**97.94**
Confidence-weighted ensemble	92.44	99.31	92.05	97.92
Gated fusion (proposed)	92.25	99.29	91.79	97.89

Noise is where models differ (55.1 to 92.5 %). Blur is saturated (98.3 to 99.5 %) because those ladders are mild at 64 px, which inflates the averaged accuracy of every model. Fog and contrast at severity 5 stay hard: the best fog accuracy is 72.7 % (SmallCNN) and the best contrast accuracy is 81.2 % (fusion).

6.3 Component ablation (E3) — RQ2

Mixing adds 4.59 points, JSD consistency 1.06, the Fourier/PRIME-style primitives 4.28, fusion 1.67 and TENT 0.18. Clean accuracy stays between 98.8 and 99.5 % throughout, so robustness was not bought with clean accuracy. The first and third steps are large relative to the interval; the JSD step (1.06) and the TENT step are within or near it and should be treated as suggestive.

Table 5. Component ablation on ResNet-18 (%). Each row adds one component to the row above.

Configuration	Clean acc.	Corr. acc.	mCE	Rel. mCE	Corr. F1	Corr. ECE
ResNet-18 baseline	98.80	83.63	258.10	328.00	81.66	9.86
+ AugMix mixing	99.35	88.22	169.00	222.00	87.04	9.61
+ JSD consistency	99.45	89.28	150.10	192.50	88.33	9.20
+ Fourier / PRIME ops	99.20	93.56	122.80	110.60	92.09	9.46
+ gated fusion with DeiT-S	99.50	95.23	**89.70**	105.50	94.07	9.11
+ TENT	99.50	**95.41**	90.50	116.60	94.20	9.61

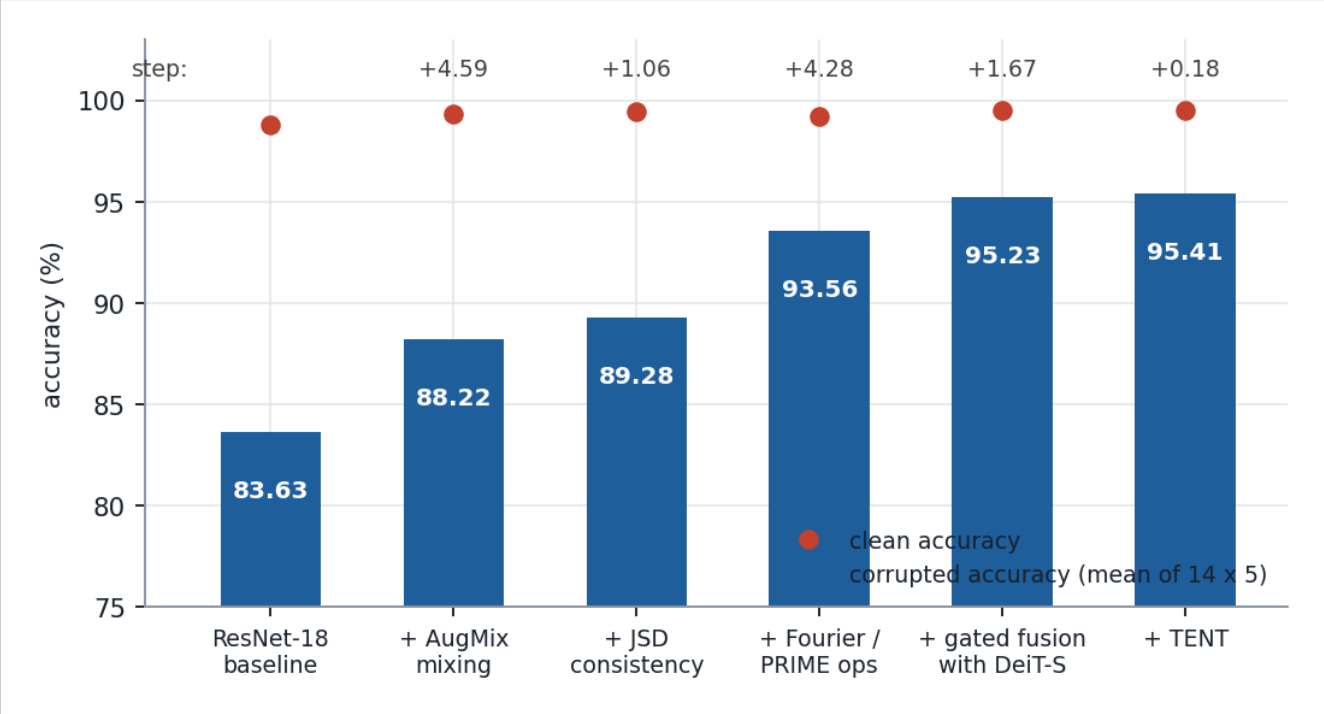


Figure 4. Corrupted accuracy along the ablation ladder (bars), clean accuracy (red dots) and the step-wise change above each bar.

6.4 Leave-category-out augmentation selection (E4)

Five ResNet-18 variants were trained. “PRIME ops only” reaches 90.69 % on noise against 71.95 % for mixing + JSD, while mixing + JSD is better on digital corruptions (97.02 against 94.78). Combining AugMix and PRIME-style ops (the proposed set) gains 2.4 points on weather and 2.1 on digital over PRIME alone and loses 1.7 on noise, for +0.95 overall. Removing the geometric ops changes the overall accuracy by −0.28 points and the digital category by +0.01, so there is no sign that their resemblance to `elastic_transform` drives the digital gain; one seed cannot settle that.

Table 6. Augmentation op-set candidates, accuracy by category (%).

Candidate	Clean	Noise	Blur	Weather	Digital	All 14
ResNet-18 (baseline)	98.80	55.06	98.32	83.43	94.25	83.63
ResNet-18 + mixing + JSD	99.45	71.95	**99.15**	87.13	**97.02**	89.28
PRIME ops only	99.10	**90.69**	98.80	87.24	94.78	92.61
AugMix + PRIME, no geometric ops	99.05	88.64	98.65	89.10	96.91	93.28
ResNet-18 (proposed training)	99.20	89.02	98.92	**89.60**	96.90	**93.56**

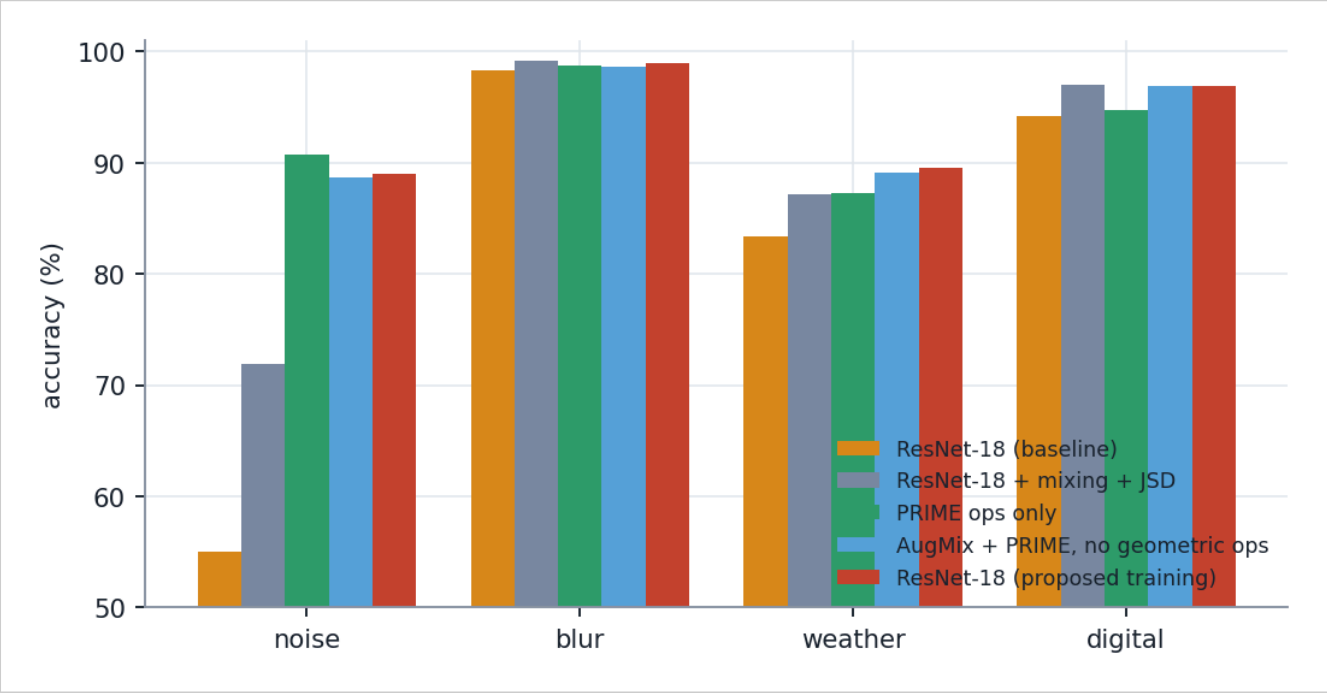


Figure 5. Accuracy by corruption category for the five ResNet-18 variants.

Selection transfers imperfectly. Choosing on noise + blur picked PRIME-only and lost 2.24 points to the best candidate on weather + digital; choosing on weather + digital picked the proposed set and lost 0.77 points on noise + blur. Both selections beat the baseline by large margins (+2.2 and +17.3 points).

Table 7. Leave-category-out selection. The gap is the oracle's accuracy minus the selected candidate's, in points.

Tuned on	Tested on	Selected	Sel. test acc.	
noise + blur	weather + digital	PRIME ops only	91.01	ResNe
weather + digital	noise + blur	ResNet-18 (proposed training)	93.97	

6.5 Gated fusion against ensembles (E2) — RQ3

The gated fusion (95.23 %, mCE 89.7) does not exceed the average ensemble (95.38 %, mCE 84.8) or the confidence-weighted ensemble (95.36 %, mCE 84.9). The accuracy differences of 0.13 to 0.15 points are inside the interval, but the mCE difference of about 5 points favours the plain average, so we find no

evidence that the gate helps. A plausible reason, which we did not test, is that both branches are near-perfect on the training set, so the gate only learns from augmented views on which they disagree.

6.6 Cross-dataset transfer (E5) — RQ5

The ordering on BTSD matches GTSRB-C: standard ResNet-18 is weakest under corruption (88.25 %, mCE 130.7) and the proposed models are strongest (fusion 94.90 %, mCE 63.3). The fusion and the confidence-weighted ensemble are within 0.05 points. Some shared classes have few images, so macro-F1 is noisy.

Table 8. Cross-dataset transfer to BTSD (%). "Shared" restricts the arg-max to the 22 classes present in both datasets; mCE uses SmallCNN on BTSD as reference.

Model	Clean	Clean (shared)	Clean F1	Corr.	Corr. (shared)	mCE	Corr. ECE
SmallCNN	94.80	95.50	91.97	90.64	91.79	100.00	22.29
ResNet-18	95.40	96.25	91.60	88.25	89.84	130.70	12.67
DeiT-S	92.30	94.70	90.57	90.91	93.01	115.60	7.07
ResNet-18 (proposed training)	95.10	95.85	91.47	92.08	93.34	97.80	9.26
DeiT-S (proposed training)	96.05	97.05	92.78	94.65	95.94	65.90	8.24
Confidence-weighted ensemble	96.25	96.90	92.53	94.85	95.84	63.90	9.57
Gated fusion (proposed)	96.20	96.90	92.67	**94.90**	**95.95**	**63.30**	9.07

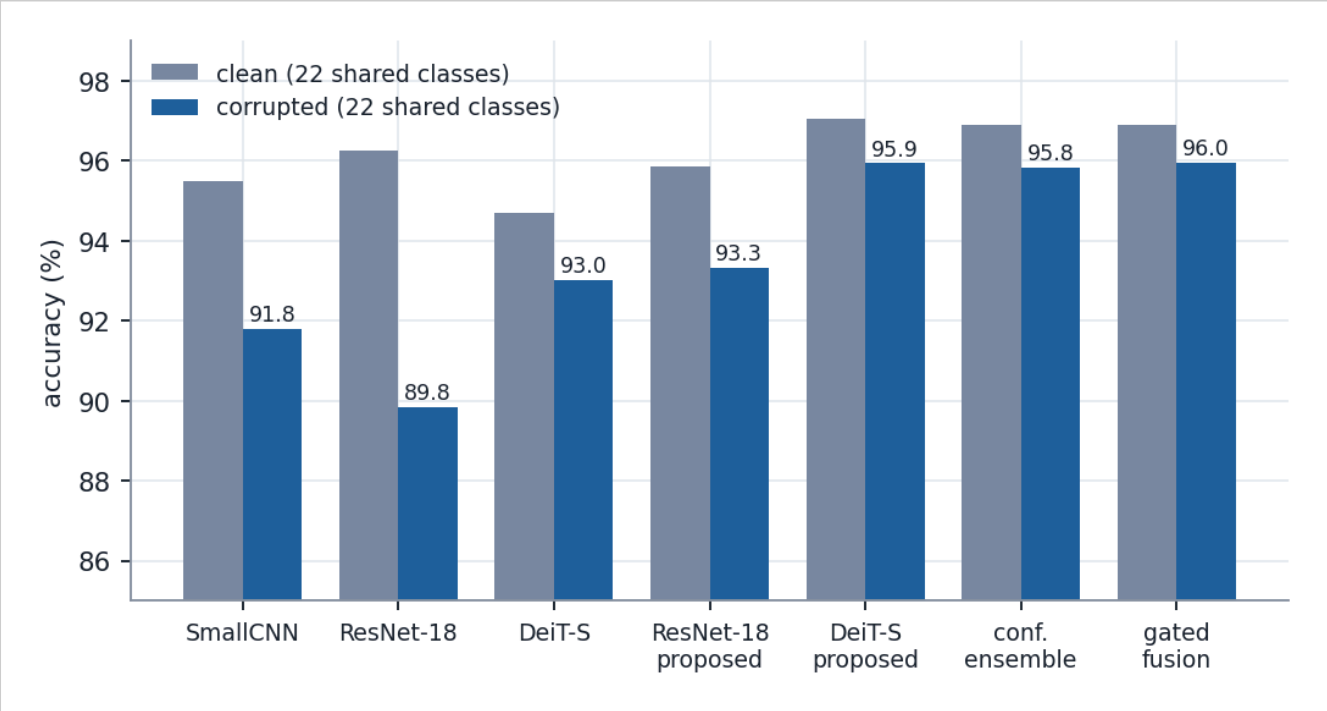


Figure 6. BTSD accuracy on the 22 shared classes, clean and corrupted.

6.7 Test-time adaptation (E6) — RQ4

- **Batch size 1 breaks BatchNorm adaptation.** SmallCNN falls from 82.7 % to 38.9 % (test-time BN) and 23.7 % (TENT); ResNet-18 from 72.8 % to 39.6 % and 17.8 %.
- **With batches of 16 or more it helps models trained without the proposed recipe.** SmallCNN gains 9.0 points at batch 16; ResNet-18 gains 7.2 to 9.9 points.
- **It adds little after the proposed training.** ResNet-18 (proposed) changes by −0.4 at batch 16 and +2.0 at batch 200; the fusion gains 0.2 to 0.55 points. This matches the null TENT step in the ablation.
- **TENT over test-time BN:** at most +0.5 points. Streams are short (500 images), which favours the statistics update over the entropy step.
- The fusion at batch size 1 loses only 0.3 points (91.86 to 91.57). A likely reason is that its DeiT branch has no BatchNorm; we did not test this separately.

Table 9. Severity-5 accuracy (%) against test batch size. The bs 1 column uses 100 images per set and the others the same 500, so bs 1 is not comparable with the rest.

Model	Method	bs 1	bs 16	bs 64	bs 200
SmallCNN	No adaptation	82.71	81.33	81.33	81.33
SmallCNN	Test-time BN	38.86	90.29	90.50	90.87
SmallCNN	TENT	23.71	90.33	90.53	90.89
ResNet-18	No adaptation	72.79	71.10	71.10	71.10
ResNet-18	Test-time BN	39.64	78.34	80.36	80.99
ResNet-18	TENT	17.79	78.87	80.77	81.00
ResNet-18 (proposed training)	No adaptation	89.29	87.81	87.81	87.81
ResNet-18 (proposed training)	Test-time BN	49.14	87.41	89.11	89.79
ResNet-18 (proposed training)	TENT	19.79	87.84	89.27	89.86
Gated fusion (proposed)	No adaptation	91.86	90.51	90.51	90.51
Gated fusion (proposed)	Test-time BN	91.57	90.69	90.90	91.04
Gated fusion (proposed)	TENT	91.57	90.81	91.00	91.06

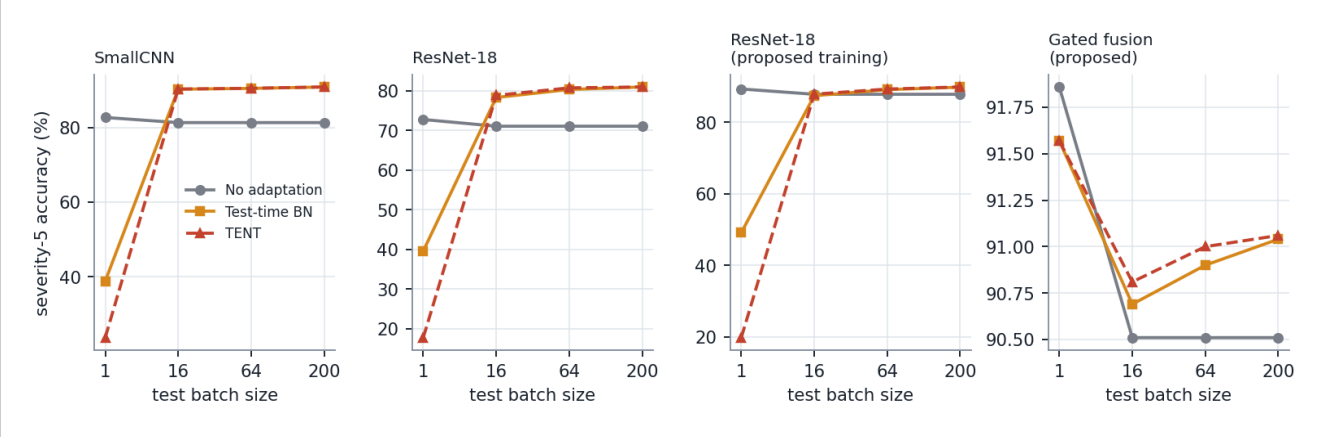


Figure 7. Severity-5 accuracy against test batch size for no adaptation, test-time BN and TENT.

6.8 Calibration — RQ6

Every model is underconfident (Figure 8). In the top bin, which holds most predictions (confidence 0.90 to 0.91), accuracy on GTSRB-C is 0.998 to 1.000 for SmallCNN and ResNet-18 and 0.989 for DeiT-S. For SmallCNN the 0.3 to 0.4 confidence bins already have 0.79 to 0.87 accuracy. Label smoothing of 0.1 over 43 classes caps the target probability near 0.90 [17, 18], so ECE is dominated by the gap between about 0.91 confidence and near-100 % accuracy. The 8 to 17 % ECE values therefore do not indicate overconfidence, and post-hoc temperature scaling [16] would probably lower them sharply; we did not apply it. Because the tables average per-set ECE while the diagrams pool the 70 sets, recomputing from the pooled bins gives slightly different values (7.65 % against 8.11 % for DeiT-S).

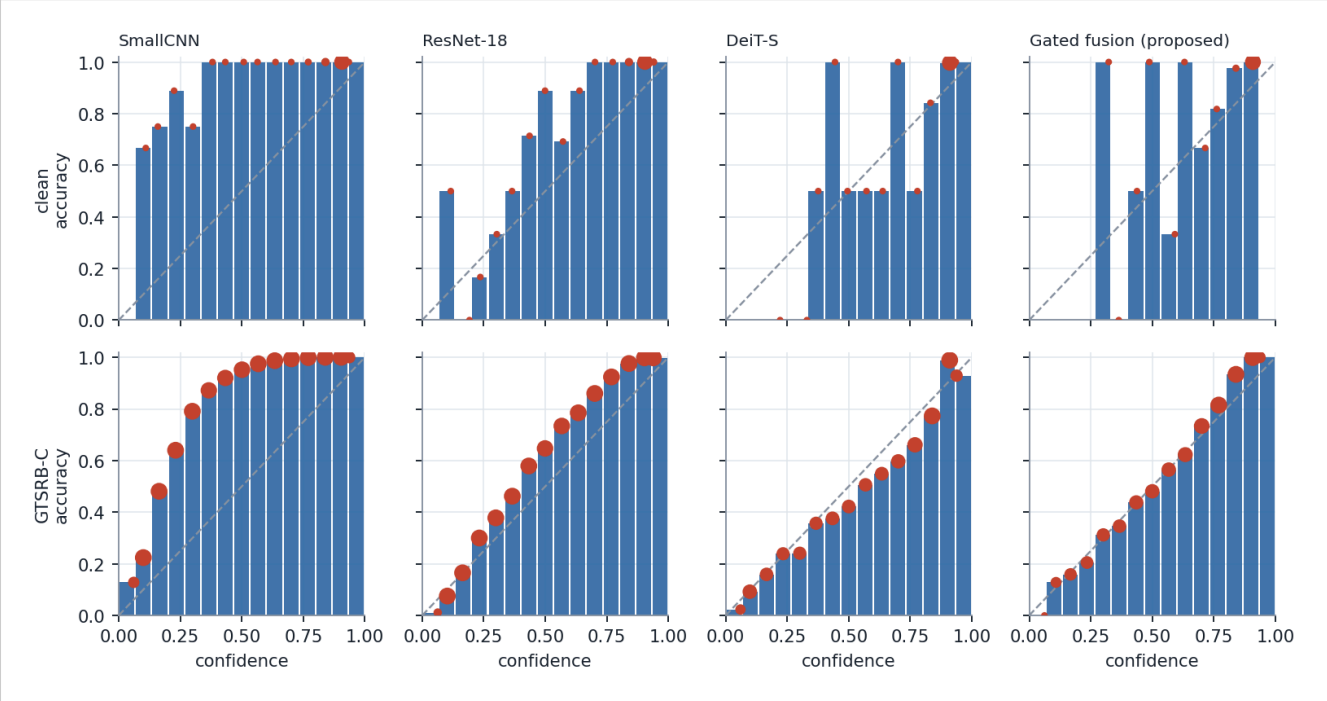


Figure 8. Reliability diagrams for clean data and GTSRB-C pooled over all 70 sets. Bars: accuracy per confidence bin; red dots: mean confidence against accuracy, sized by bin count. Points above the dashed diagonal are underconfident.

7 Discussion

The pilot supports a clear, practical conclusion: robustness comes mainly from the training recipe, not from the architecture or from inference-time tricks. The recipe is cheap, costs no clean accuracy here, and its two largest steps are exactly the mixing and the spectral/spatial primitives. The learned fusion and TENT, the two components that add complexity at inference time, earned little at this scale.

Table 11. Answers to the research questions.

	Finding	Strength of evidence
RQ1	Yes, by a large margin: ResNet-18 83.63 % → 93.56 % corrupted accuracy (+9.93); DeiT-S +1.55.	Strong: far outside the conservative interval; single seed.
RQ2	Mixing (+4.59) and the Fourier/PRIME-style primitives (+4.28) dominate; JSD adds +1.06.	Large steps strong; JSD step suggestive.
RQ3	No. The learned gate does not beat the plain average (95.23 % against 95.38 %; mCE 89.7 against 84.8).	No evidence of benefit; accuracy gap within noise.
RQ4	Helps models without robust training at batch ≥ 16 (+7 to +10 points); small after proposed training; collapses at batch 1. TENT adds at most 0.5 over test-time BN.	Consistent across four models; short streams.
RQ5	Ordering preserved on BTSD; fusion mCE 63.3, ResNet-18 baseline 88.25 % corrupted.	One dataset, 22 classes, one seed.
RQ6	All models underconfident; high ECE is mostly a label-smoothing effect.	Directly visible in reliability bins.

Comparison with published work. On ImageNet-C, AugMix reports 65.3 mCE with a ResNet-50 [4] and PRIME reports 57.5 [5]. Our mCE values are normalised by a different model on a different benchmark and cannot be placed on that scale. Clean accuracy of 98.8 to 99.6 % is in the band of published GTSRB results [2], but a 2,000-image subset cannot resolve differences at that level. We found no published score table for a comparable GTSRB corruption benchmark, so we make no ranking claim against other work.

8 Limitations and threats to validity

- **One seed, 40 % schedules.** No standard deviations or significance tests; absolute accuracies are likely below full-schedule values, and the relative effect of three-view training may differ.
- **Small, repeated test set.** The 140,000 corrupted samples are 2,000 images under 70 corruptions, so the effective sample size is closer to 2,000.
- **Validation corruptions are relatives of test corruptions** (speckle ~ Gaussian/shot noise, Gaussian blur ~ defocus blur), so model selection is not fully independent of the noise and blur categories.
- **Operation overlap.** Geometric operations resemble `elastic_transform`, the random spectral filter resembles mild blur, and colour curves and jitter resemble brightness and contrast. Only the first is tested (Section 6.4).
- **Mild ladders.** JPEG, pixelate, defocus and zoom blur barely hurt at 64 px.
- **Normalisation.** mCE depends on the SmallCNN reference.
- **Fourier sensitivity map.** It was produced but is not interpreted here: models trained with Fourier-basis noise are not independent of a test that perturbs along such bases.

- **Scope.** 64 px images, two datasets, 22 shared classes, synthetic corruptions, one GPU model and library stack.

9 Conclusion and future work

On a 64 px GTSRB corruption benchmark, a clean-trained ResNet-18 loses about 15 points of accuracy under corruption, and a mixed-augmentation recipe with PRIME-style primitives recovers about two thirds of that loss at no measurable cost in clean accuracy. A learned fusion gate and entropy-based test-time adaptation did not add measurable benefit over a plain average and over test-time BN. The next step is the full experiment: three seeds, full schedules and the complete 12,630-image test set, which would give standard deviations, allow significance testing, and settle the small differences the pilot cannot resolve. Further work should store per-image predictions for paired tests, evaluate temperature scaling, test the geometric-operation overlap with more seeds, and add CTSD.

Reproducibility

The corruption sets are generated from fixed seeds and cuDNN is set to deterministic kernels. Pilot wall-clock times on the T4: corrupted GTSRB and BTSD sets 9.2 and 8.7 minutes, experiments 1 to 6 respectively 31.1, 6.4, 11.2, 12.8, 5.2 and 3.8 minutes, figures 2.0 minutes (generation overlapped experiment 1). Result tables are available as CSV, text and LaTeX for every experiment.

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Appendix A Severity-5 accuracy per corruption

Table 10. Severity-5 accuracy per corruption and model (%).

Corruption	SmallCNN	ResNet-18	DeiT-S	ResNet-18 prop.	DeiT-S prop.	Avg ens.	Conf.
gaussian noise	59.90	29.80	74.40	78.50	80.70	83.00	82
impulse noise	48.90	17.00	66.50	64.20	73.00	73.80	73
shot noise	75.80	31.10	87.10	91.90	91.30	93.30	93
motion blur	98.90	**97.90**	97.80	98.20	99.00	99.10	99
zoom blur	99.50	97.50	**99.10**	98.80	**99.30**	**99.40**	**99**
defocus blur	99.00	95.90	97.20	98.50	98.70	98.80	98
snow	83.50	70.30	88.80	86.60	90.50	91.40	91
fog	72.70	63.10	66.00	65.60	67.90	69.20	69
frost	78.80	71.90	85.90	82.80	86.90	87.10	87
brightness	94.20	90.30	92.50	94.00	94.90	95.30	95
contrast	38.40	55.20	70.40	71.50	77.40	80.30	80
pixelate	**99.60**	96.90	99.00	**99.10**	99.20	99.30	99
elastic transform	98.30	90.80	96.50	96.70	97.20	98.00	98
jpeg compression	98.20	93.50	96.90	97.80	98.00	98.60	98

Appendix B Result files

Per-experiment tables (`.csv`, `.txt`, booktabs `.tex`) for E1 to E6 and per-severity and per-category breakdowns, plus vector figures for severity curves, reliability diagrams, the Fourier sensitivity map, Grad-CAM panels and confusion matrices, were written to `results_pilot/` on Google Drive. The Fourier map, Grad-CAM panels and confusion matrices are not reproduced in this paper.